



ASISTENTE EDUCATIVO INCLUSIVO

Modo de adaptación: **Lectura fácil**

TEXTO ORIGINAL

Integrating renewable energy is critical for maintaining grid stability while striving towards global climate goals. Concentrating solar power tower plants offer a promising solution but their competitiveness is currently hindered by high operational costs, limited data availability and slow adoption of emerging technologies. To address these barriers, here we introduce PAINT, a FAIR (findable, accessible, interoperable and reusable) open-access database for operational solar tower plant data. PAINT provides 849 GB of high-resolution data collected over multiple years from the Jülich solar tower plant, including heliostat properties, calibration and deflectometry measurements and fine-grained weather data. The database is organized using the SpatioTemporal Asset Catalog metadata specification and supports the development of digital twins, artificial intelligence-based calibration methods, predictive maintenance and improved solar flux prediction. We also introduce standardized benchmarks to promote reproducibility and fair comparisons. Providing access to high-quality data, PAINT enables broader participation in solar research, accelerates innovation and facilitates data-driven solutions in solar tower power plant research.

TEXTO ADAPTADO

PAINT is a FAIR (findable, accessible, interoperable and reusable) open-access database for operational solar tower plant data. It provides 849 GB of high-resolution data collected over multiple years from the Jülich solar tower plant. It supports the development of digital twins, artificial intelligence-based calibration methods, predictive maintenance and improved solar flux prediction.